CS & IT ENGINEERING



Algorithms

Graph Algorithms

Lecture No.- 03



By- Aditya sir

Recap of Previous Lecture



Topic DFS on Directed graphs

Topic DAG

Topics to be Covered



Topic Application of DFS & BFS

Topic Connected components

ABOUT ME

Hello, I'm Aditya.

1. Represented college as the first Google DSC Ambassador.
2. The only student from the batch to secure an internship at Amazon.
3. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis8. Completed my Masters with an overall GPA of 9.36/108. Joined Dream11 as a Data Scientist.9. Have mentored around 10k working professions in field of Data Science and Analytics10. Have been mentoring GATE aspirants to secure a great rank in limited time.11. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.



By- Aditya sir



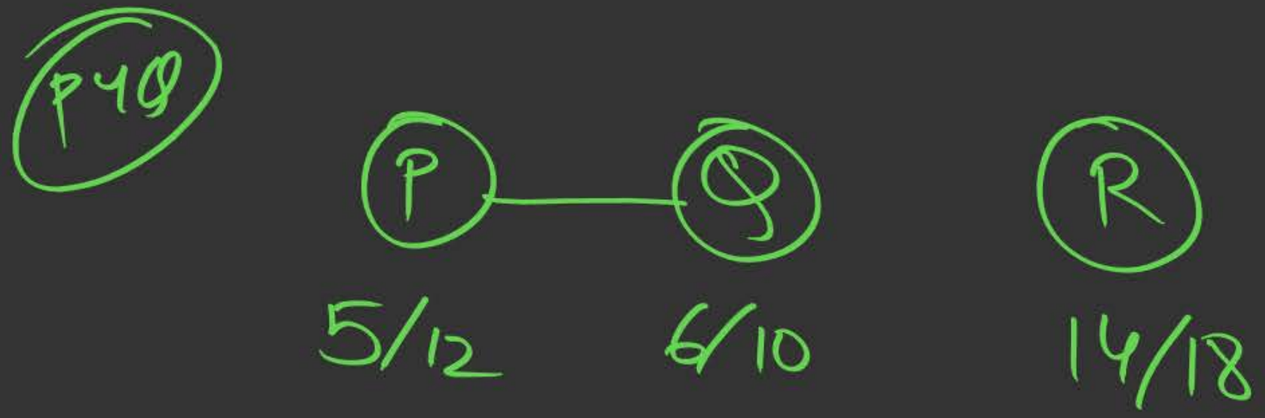
Topic : Graph Algorithms

Time complexity:-

1. Time complexity of both DFS & BFS dependents on how the graph is represented.
   (a) Adjacency matrix: $O(n^2)$
   (b) Adjacency list: $O(n + e)$

n = number of vertices

e = number of edges



Ans :- D

2 components





Topic : Graph Algorithms

Application of DFS and BFS:

1. Both DFS and BFS can be used to determine whether the graph is connected on disconnected.
2. Both DFS BFS can be ~~need~~ used to check/ detect whether the two vertices u and u connected or not.
3. Cycle detection: Both DFS and BFS can be used to detect the presence of a cycle in the given graph.



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4. DFS can also be used to determine:

   (a) Connected components in a graph→ [undirected graph]

   (b) Strongly connected components → [directed graphs]

   (c) Bi- connected components (BCC)

   (d) Articulation points/cut vertex.

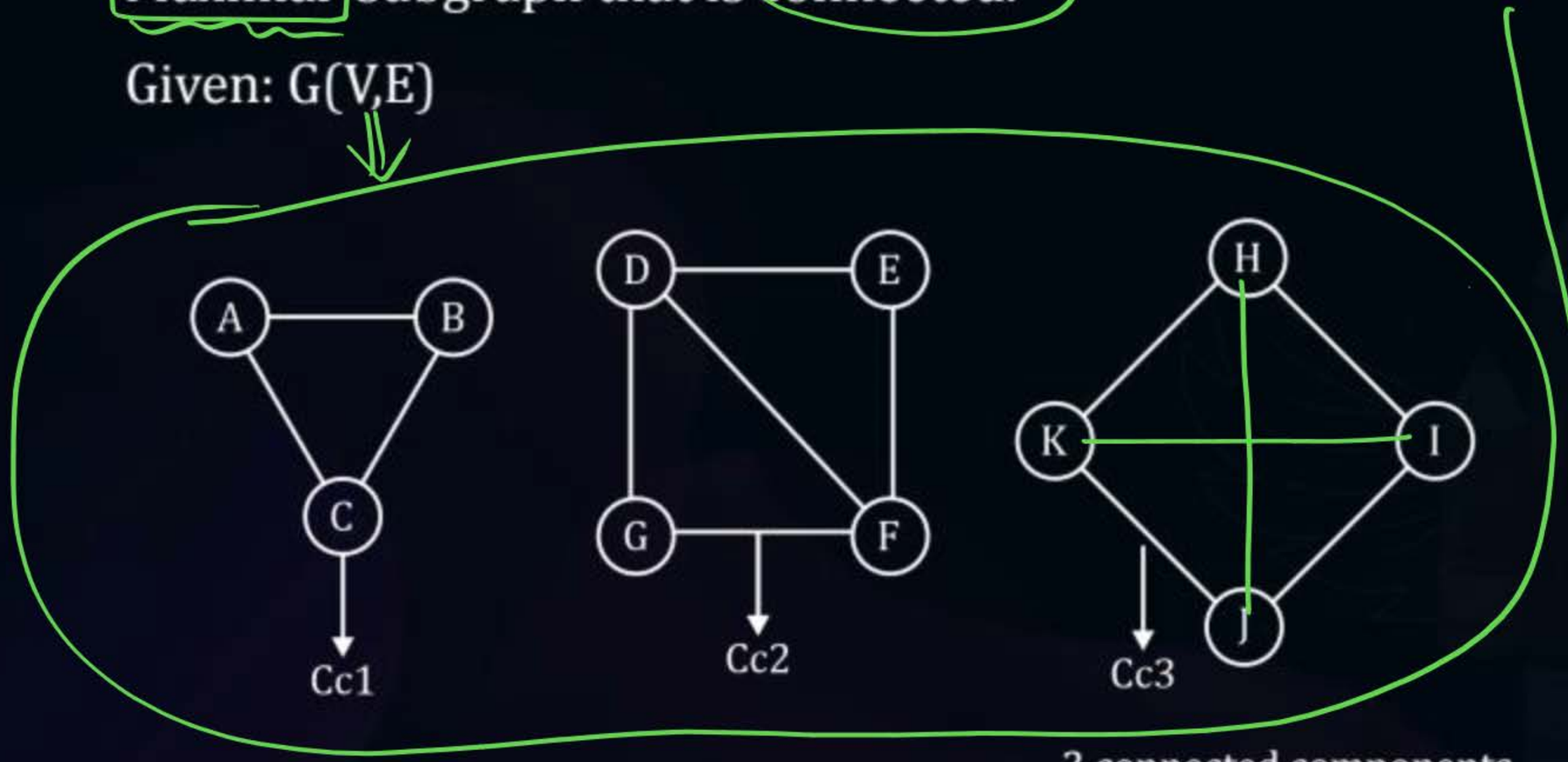


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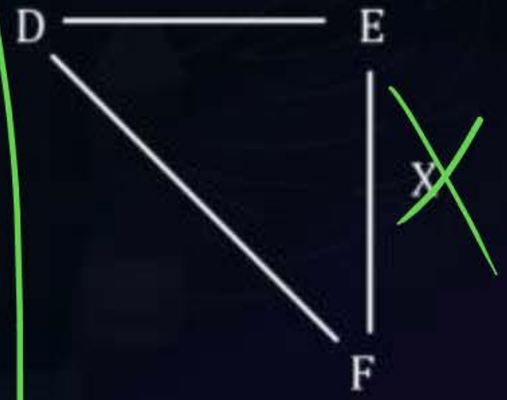
1. Connected components (CC) → [undirected graphs]

   Maximal subgraph that is connected.

   Given: G(V,E)



3 connected components

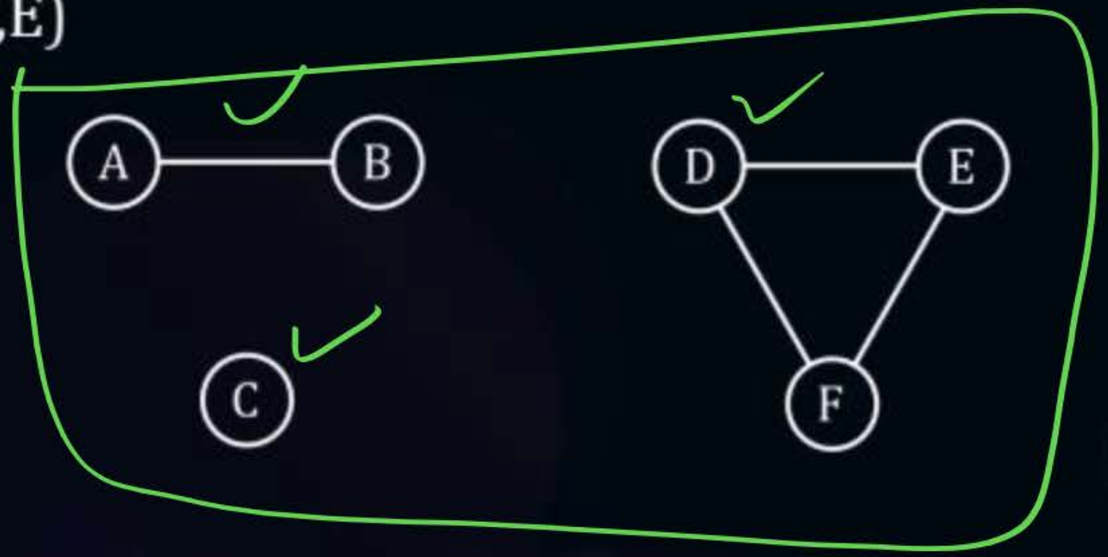




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E.g. 2.

G(V,E)



3 connected components

Single vertex can also be connected components



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2. Strongly connected components (SCC)

(Directed Graph)

V= [set of all vertices in G]

Definition:-

1. Two nodes 'u' and 'v' of directed graph are connected, if there is a path from 'u' to 'v' and there is also a path 'v' to 'u'.

u ⇝ v

2. The relation, partitions the vertex set V into disjoint set which are known as strongly connected components (SCC).

$S_1 \cap S_2 = \phi$



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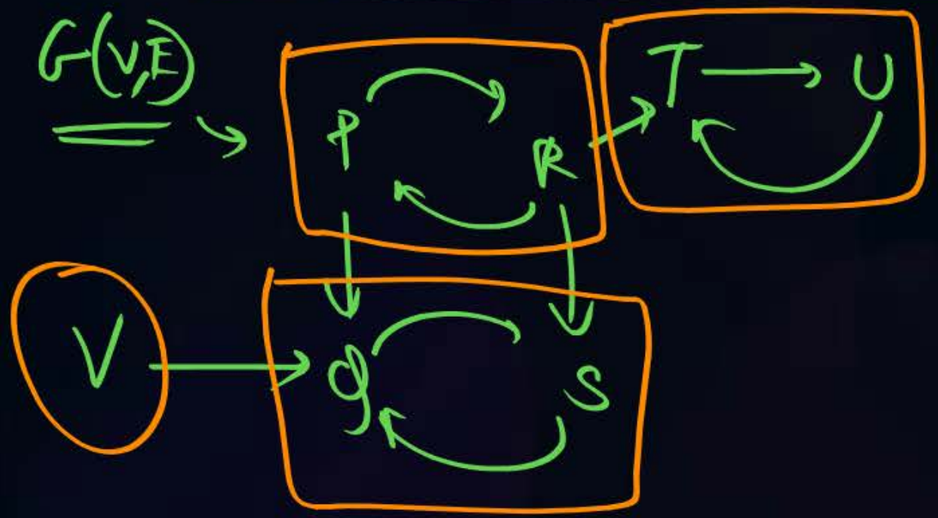
E.g.3. Given: (V,E)

Strongly connected components

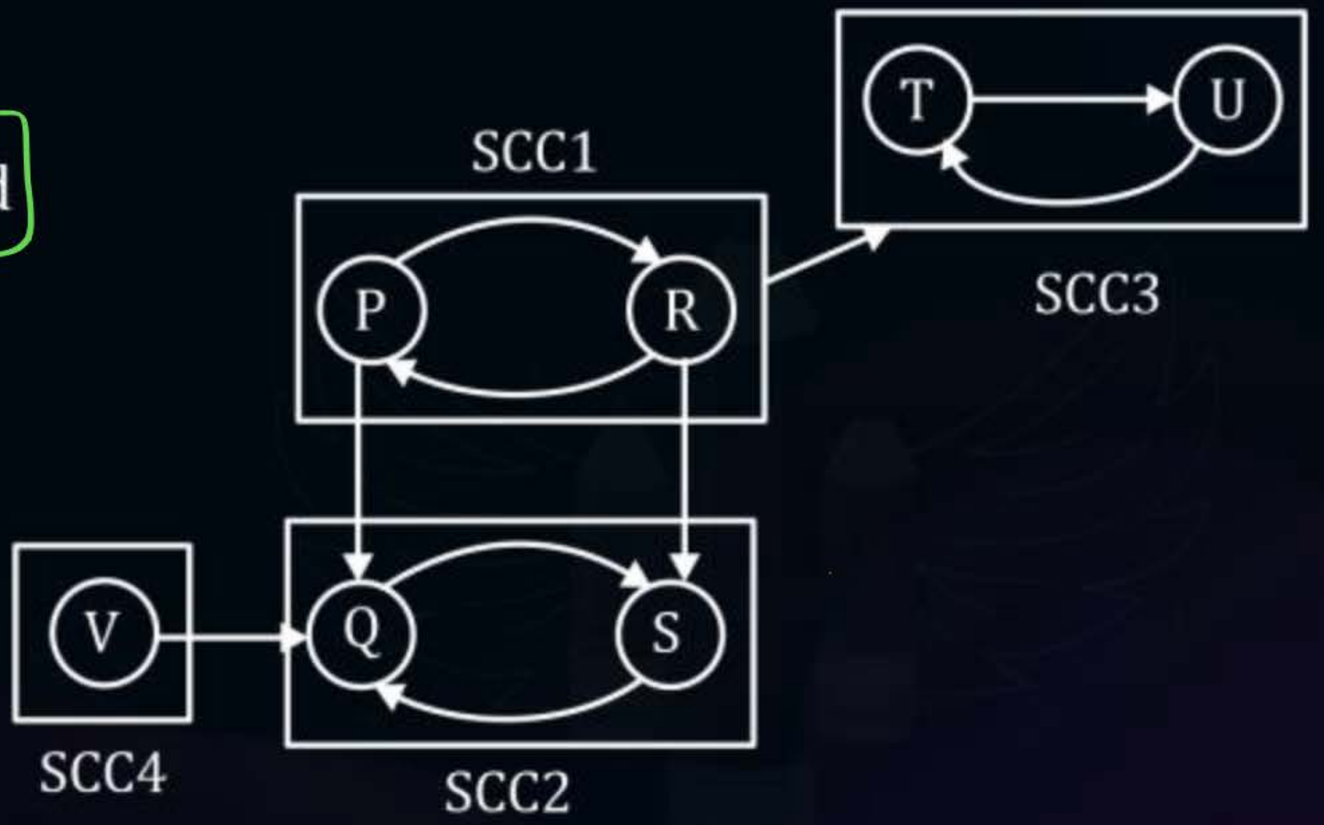
↓

CC

maximal subgraph that is connected



4 SCCs





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In above eg. There are 4 strongly connected components.

SCC1	SCC2	SCC3	SCC4
{P,R}	{Q,S}	{T,u}	{v}

Combine them you get 'V'

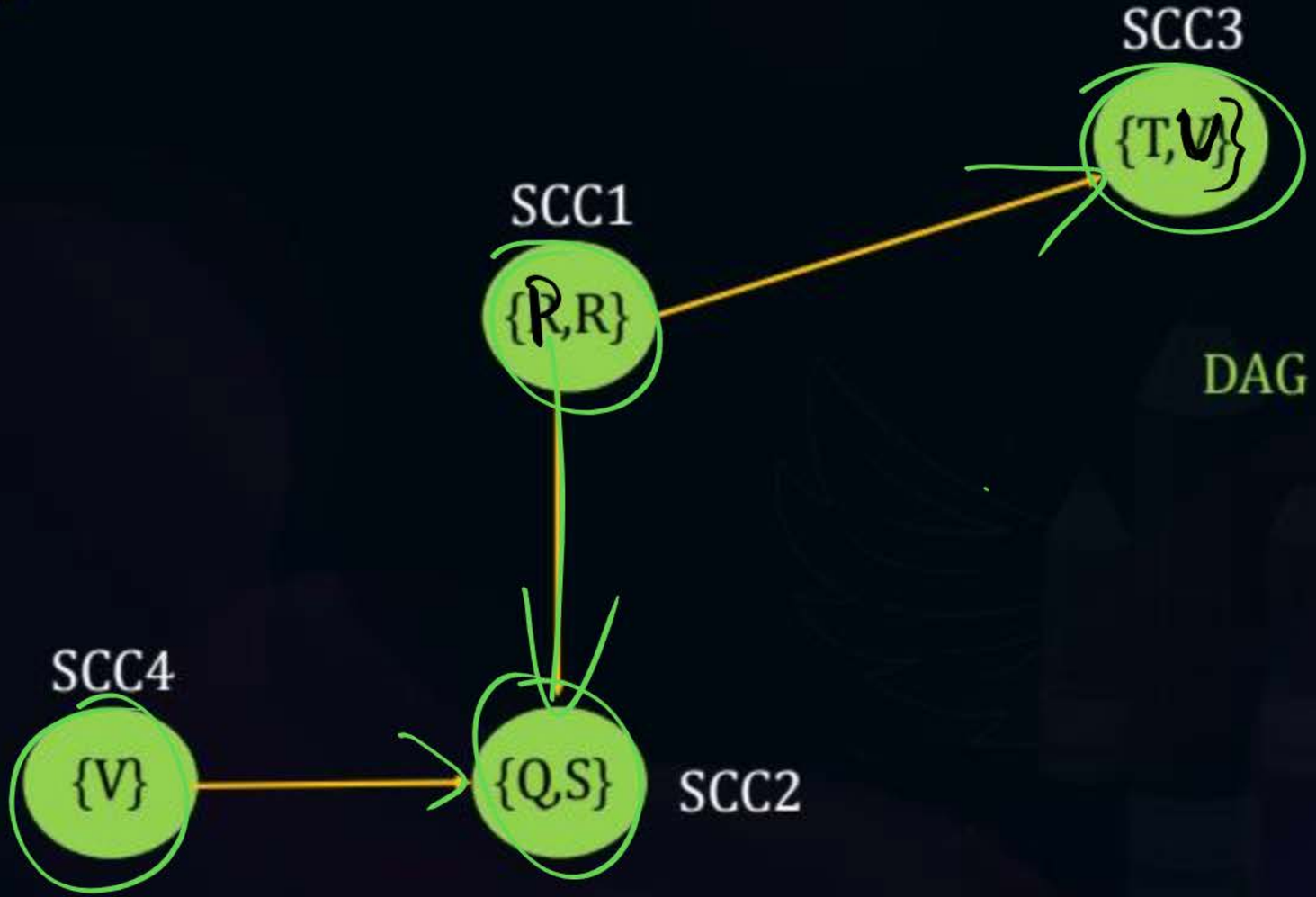
'V' → {P,Q,R,S,T,U,V}

disjoint

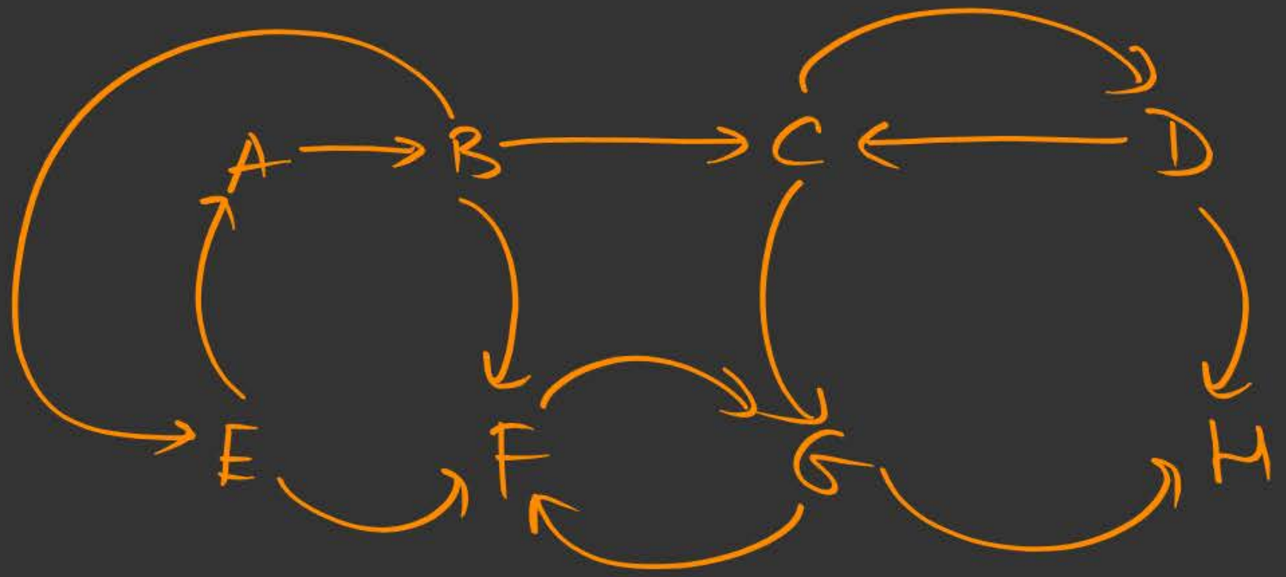


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Validate property:-



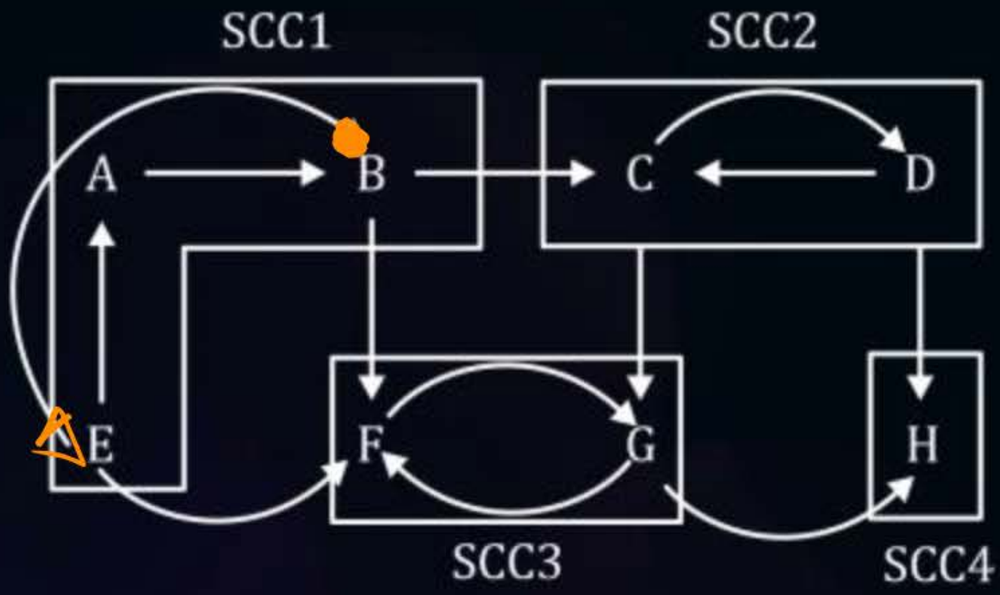
(8)





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#Q. How many strongly connected components are there in the given graph G?



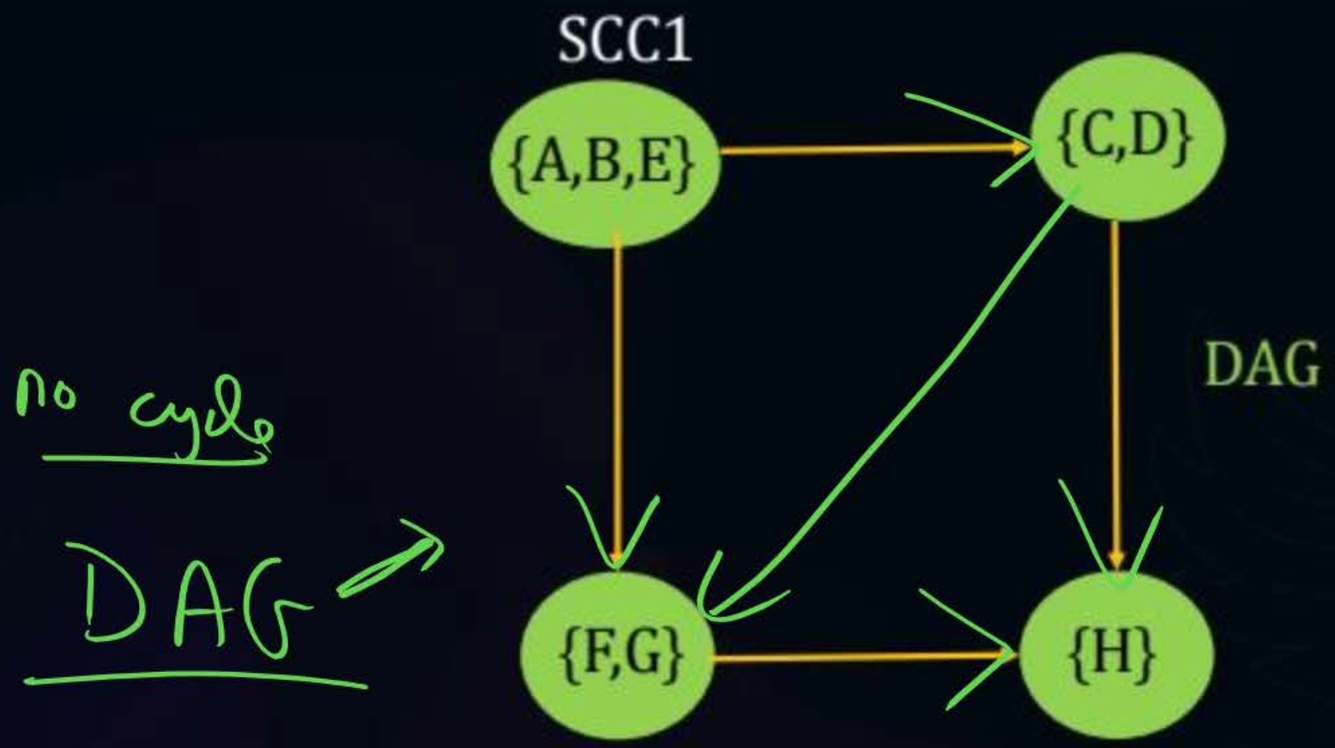
SSC 1 → {A, B, E}
SSC 2 → {C, D}
SSC 3 → {F, G}
SSC 4 → {H}
'V' = {A,B,C,D,E,F,G,H}

Disjoint



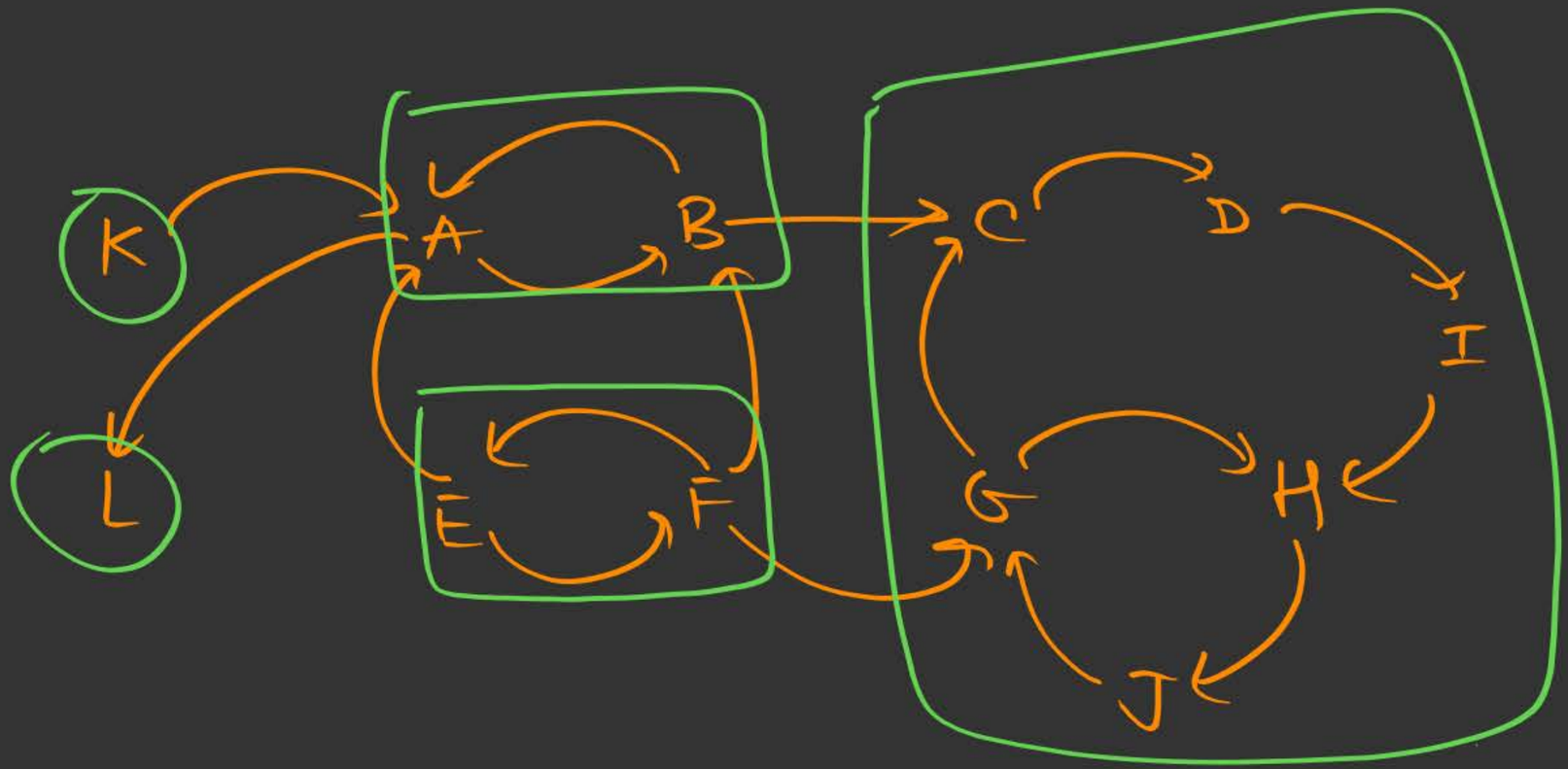
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Property:-1



no cycle

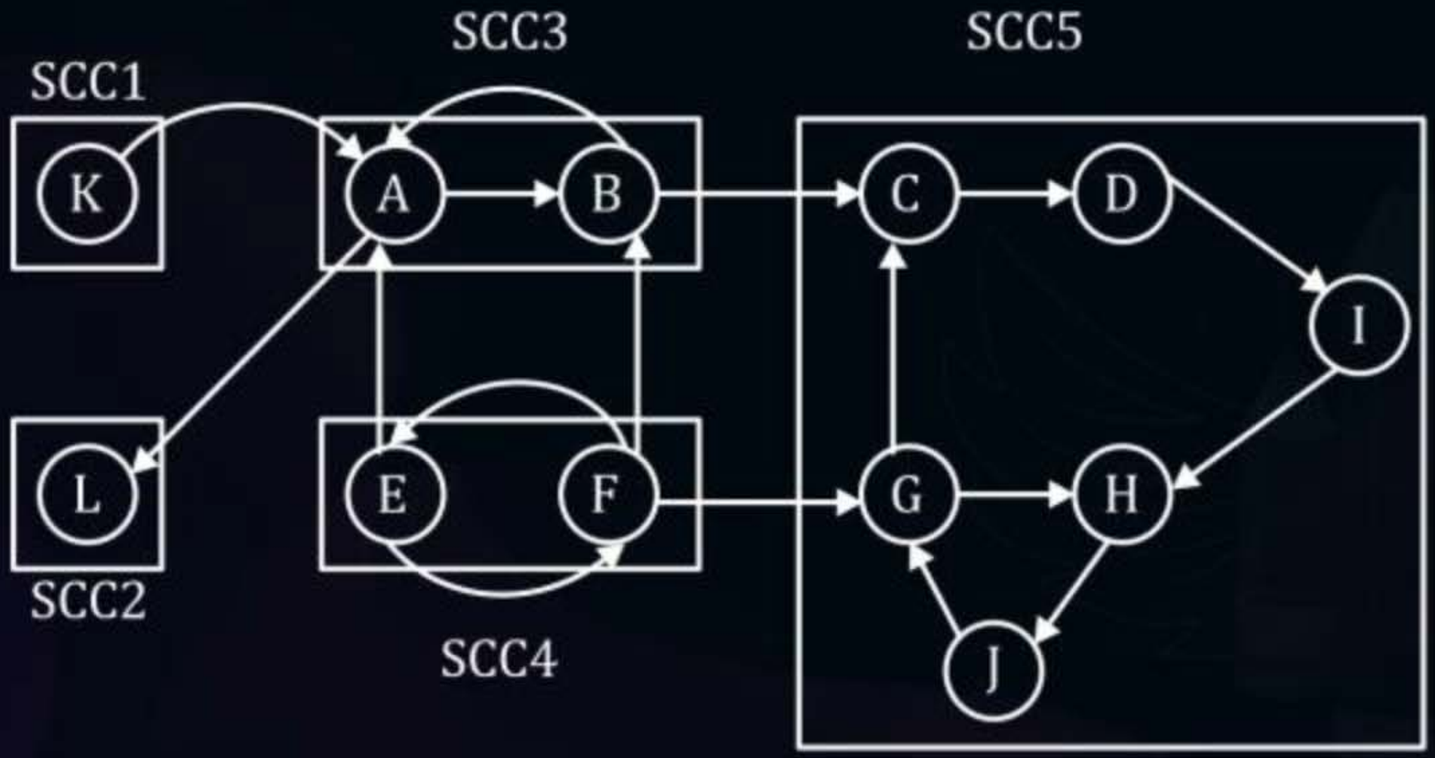
DAG


K
L
A
B
E
F
C
D
I
G
H
J



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#Q. How many strongly connected components are present in given graph and what are those?





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SSC → CC: Maximal subgraph that is connected

SSC 1 → {K}

SSC 2 → {L}

SSC 3 → {A,B}

SSC 4 → {E, F}

SSC 5 → {C, D, G, H, I, J }

Disjoint

Combine the get 'v'

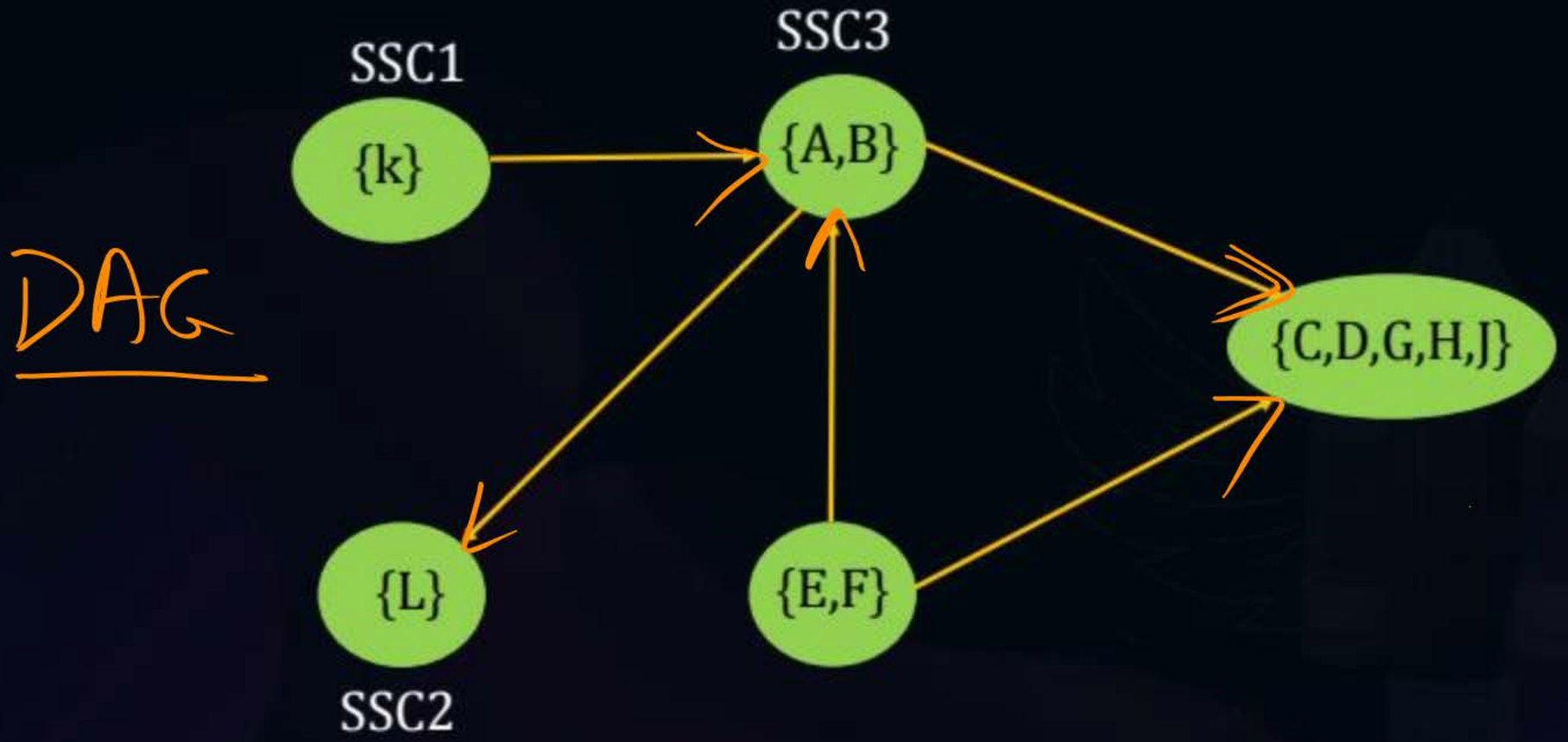




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Property 1

DAG



DAG



Topic : Graph Algorithms

SCC

Property-1: Every directed graph is a D.A. G of its strongly connected components.

Property-2 : Let C and C' be distinct strongly connected components in directed graph G = (V, E), let u, v, ∈ C and u^1, v^1 ∈C^1, suppose that there is a path u~u^1 in G, then there cannot also be a path v^1~ v in G.

Property-3: If 'C' and 'C^1' are strongly connected components of, and there is an edge from a node in C to a node in C^1, then the highest post number in C is bigger than the highest post number in C^1.

Finishing time

Finishing time



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Property 2 explanation:

In u→ u is a path then there cannot be a path from v → v (otherwise it wont be a DAG and property will be invalid.)



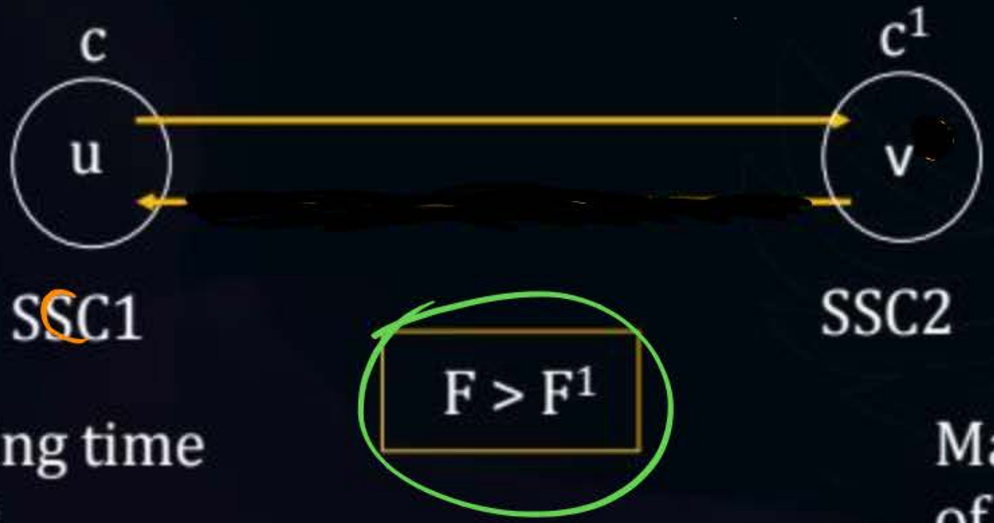


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Property-3 explanation:

The highest finishing time of c be f

Max among the finishing time of use the vertices in c



The highest finishing time of c^1 be f^1

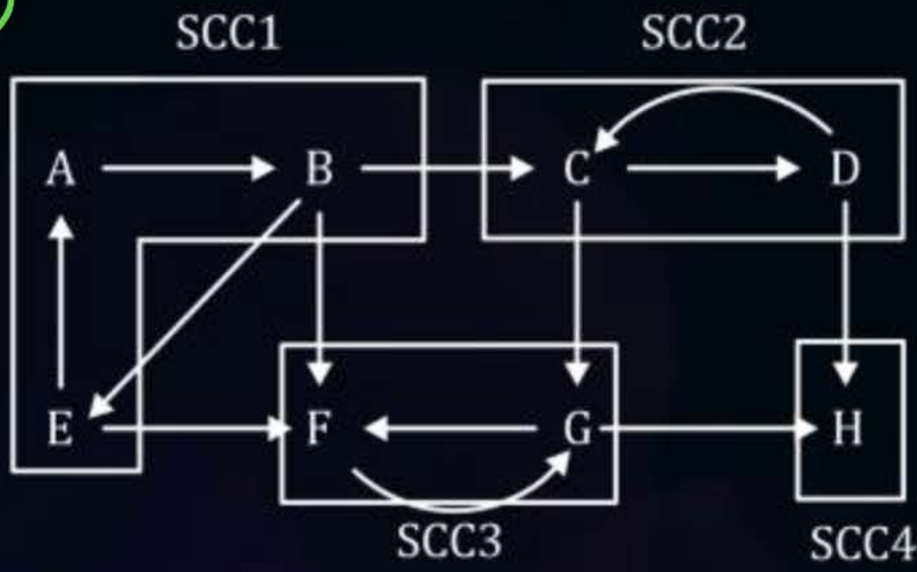
Max among the finishing time of use the vertices in c^1



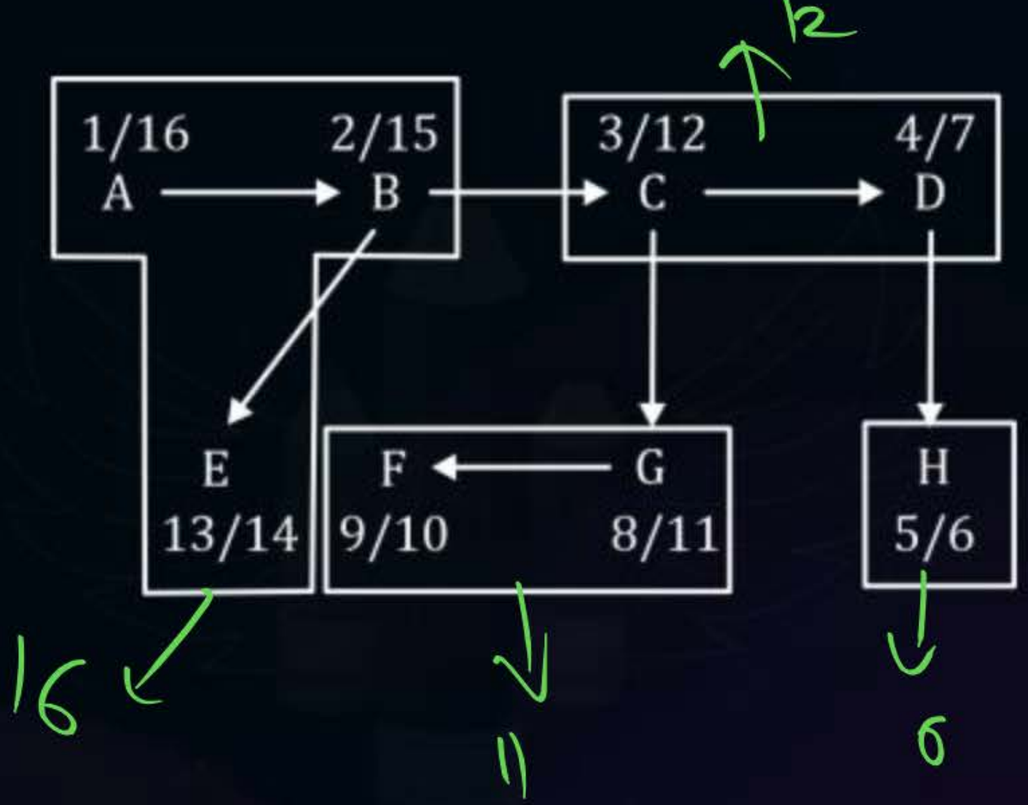
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Example to understand this property -3:

d/f



DFS Starting at vertex A.





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SSC 1→ SSC2

↓ ↓ 16 >12

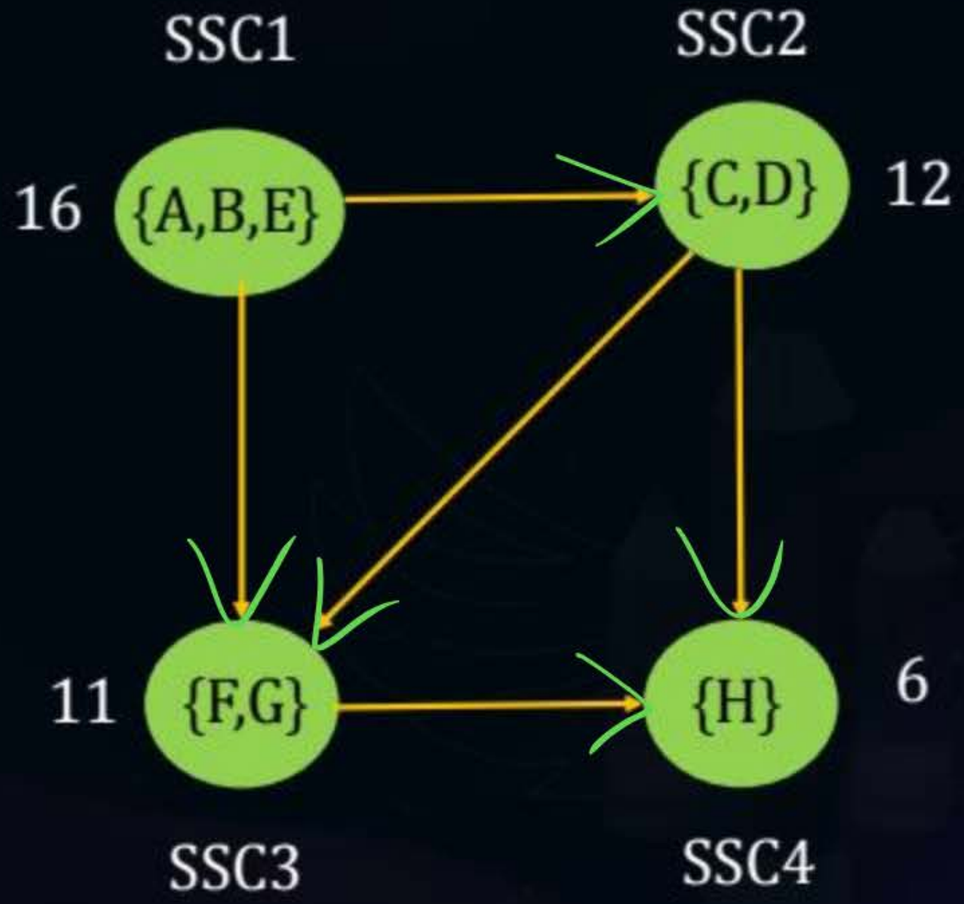
16 12

SSC 1 → SSC 3: 16 > 11

SSC 2 → SSC 3: 12 > 11

SSC 3 → SSC 4: 11 > 6

SSC 2 → SSC 4: 12 > 6



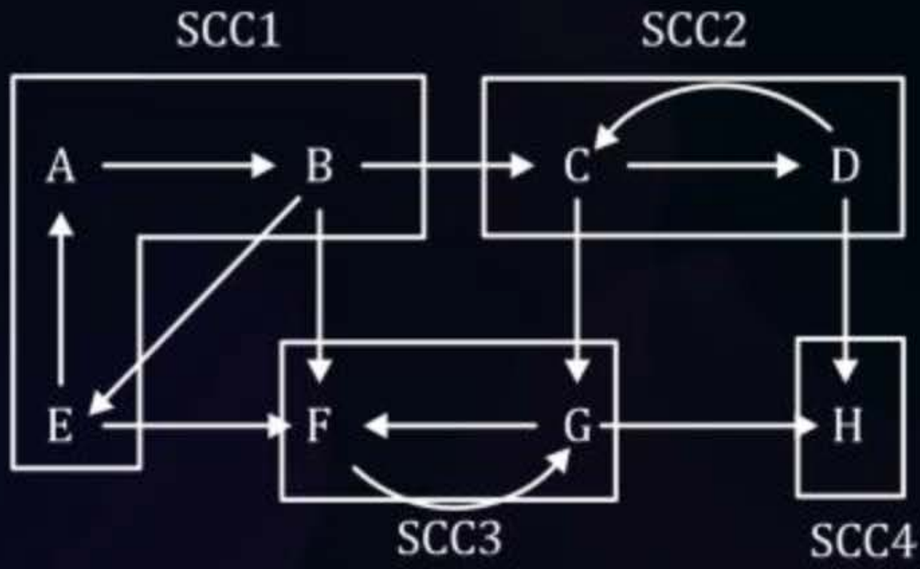


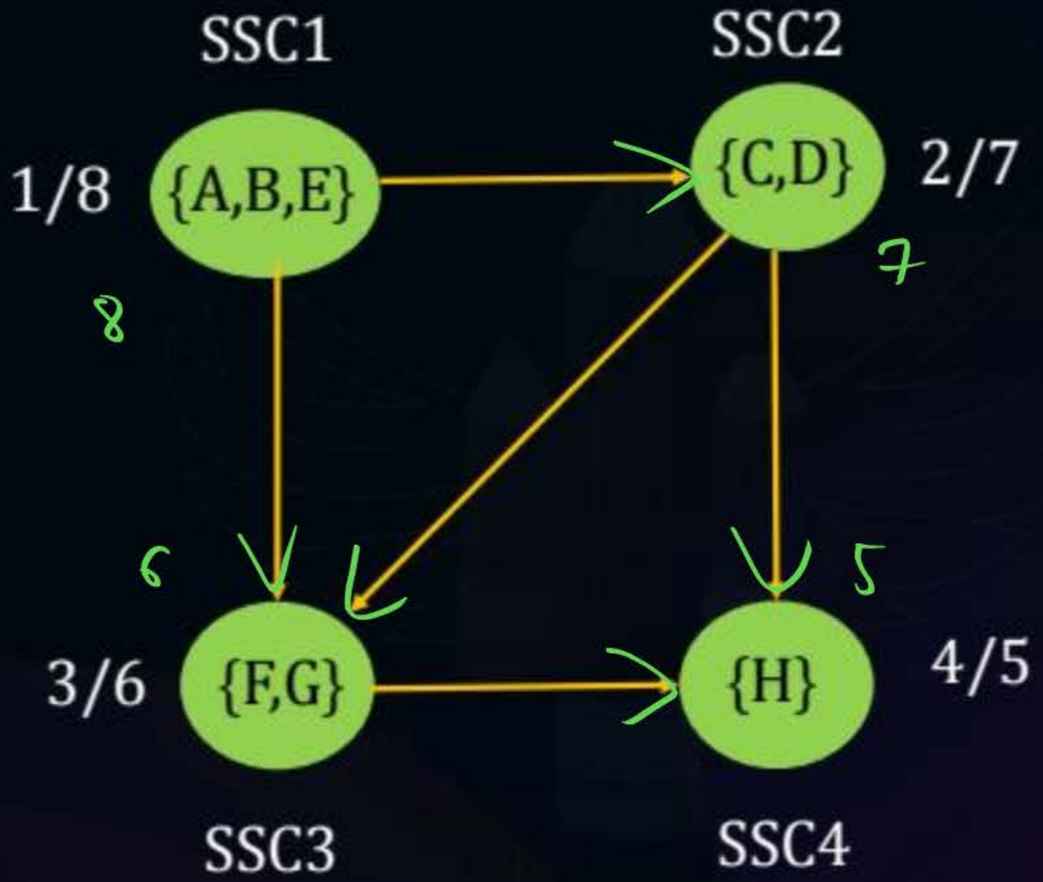
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Bonus concept for property-3 (ShortCut)

If we apply DFS on the SSC DAG, then also property 3 can be validated.

Given:







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Finishing time

SCC 1 → 8

SCC 2 → 7

SCC 3 → 6

SCC 4 → 5

1→ 2: 8 > 7

2→ 4: 7 > 5

2→ 3: 7 > 6

1→ 3: 8 > 6

3→ 4: 6 > 5

Articulation Point

/Cut vertex



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Articulation point / cut vertex

When do we call a vertex as a cut vertex / articulation point?

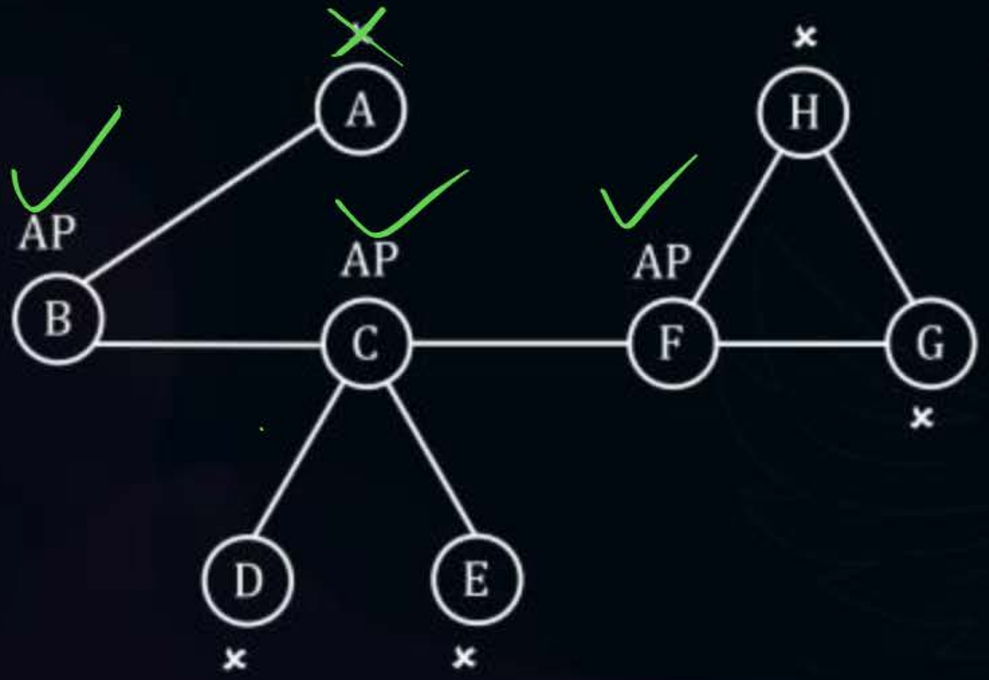
Articulation point in a graph in such a vertex, which when removed along with all of its edges from the given graph, will partition the graph into 2 or more non-empty components.



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E.g.4. How many articulation point are there in the below graph? What are those?

Given: (G(V,E))



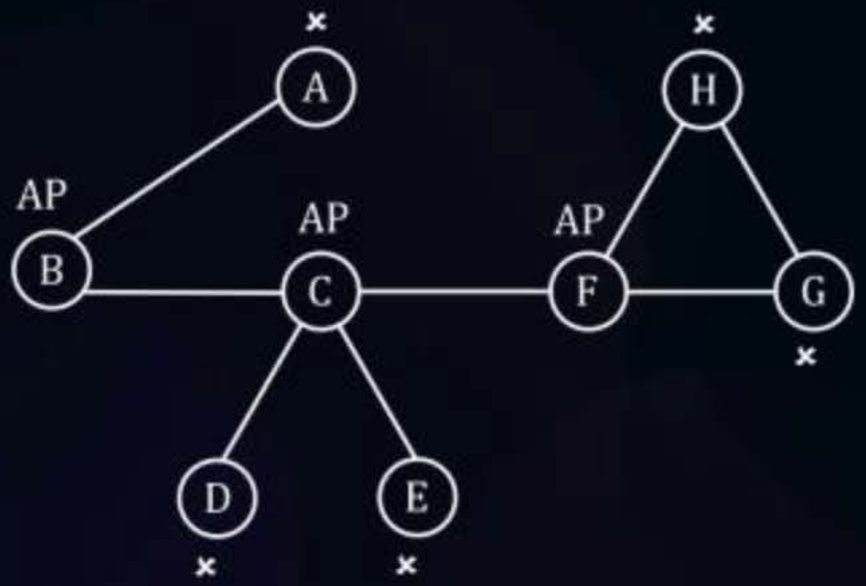
→ 3 articulation points

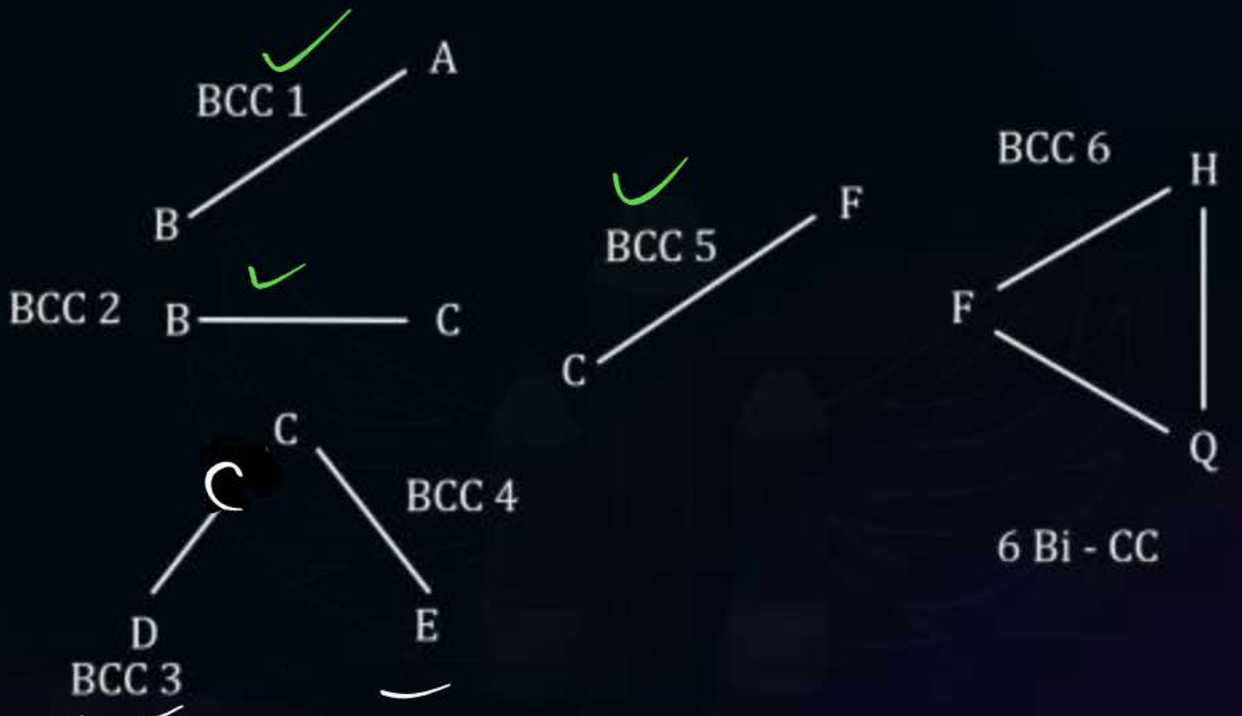
B, C and F

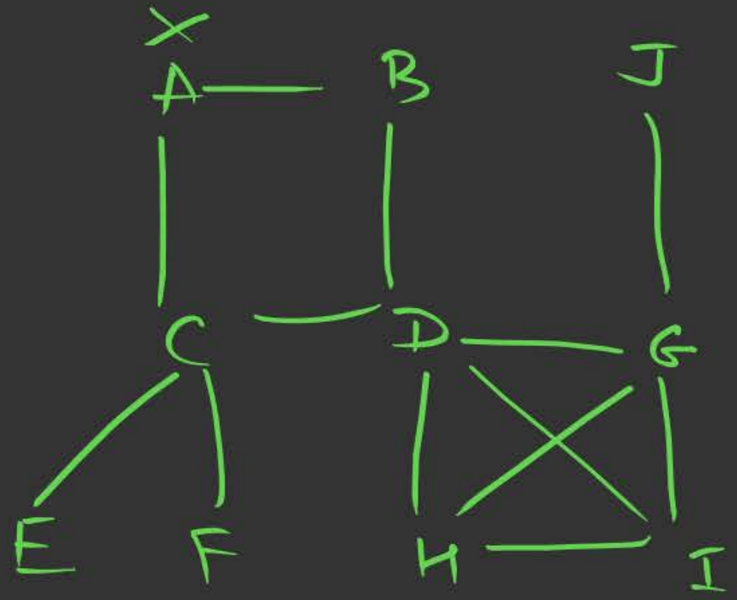


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- AS 3 Aps not a bi-connected graph:
- How many bi-connected components does it has?





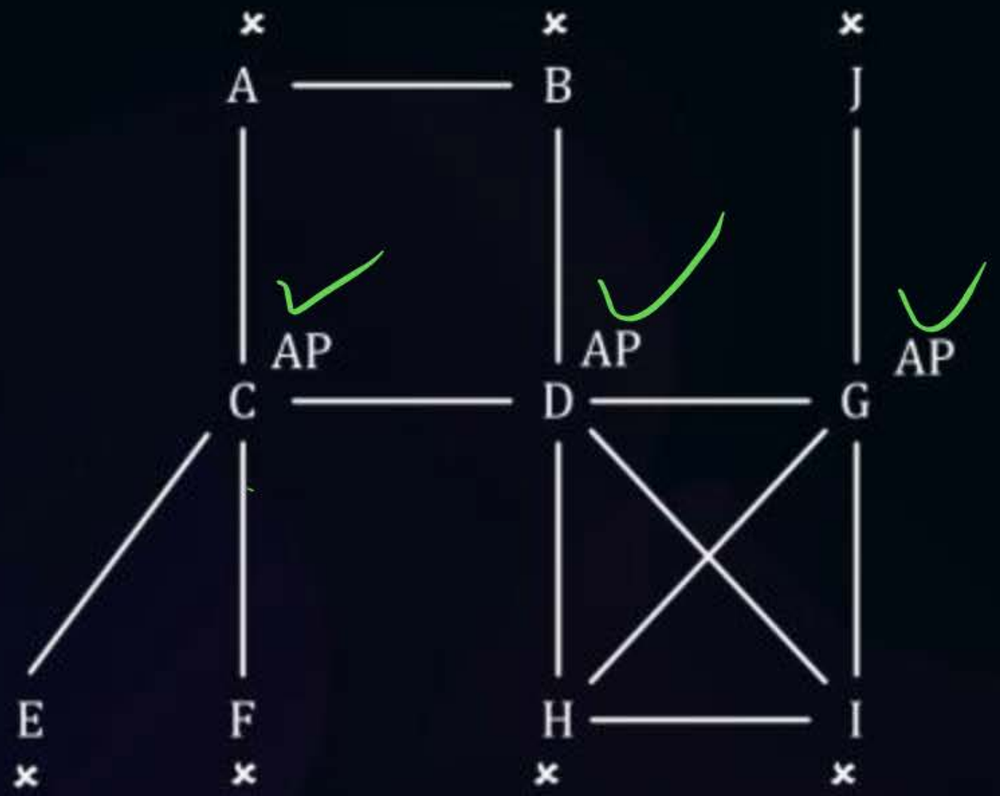

X
A
B
J
C
D
G
E
F
H
I



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18.5 %

#Q. How many articulation point and what are these?



3 Articulation Point

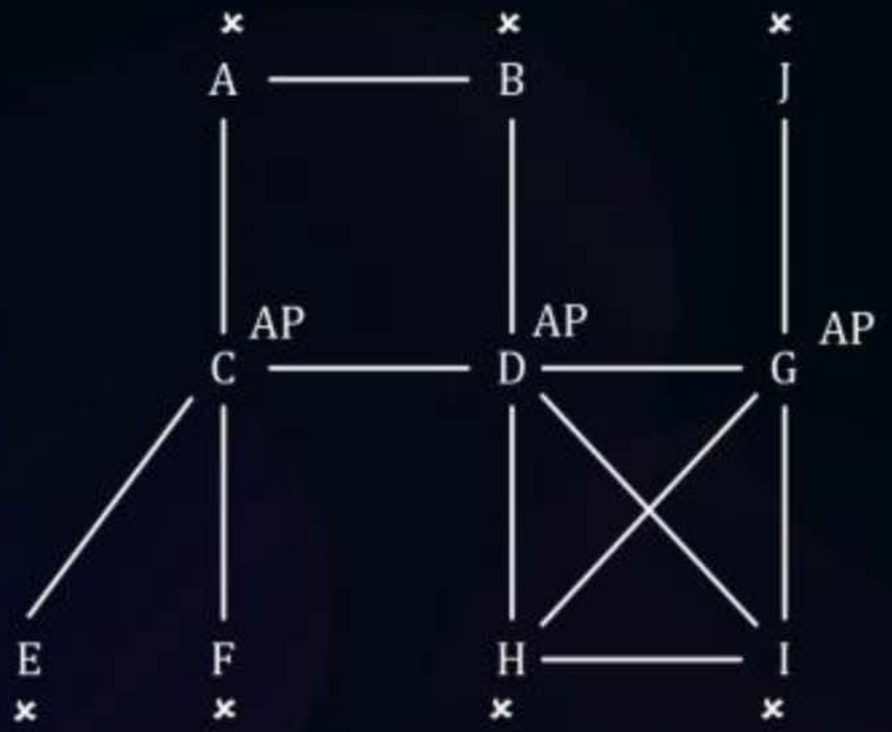
C, D, and G

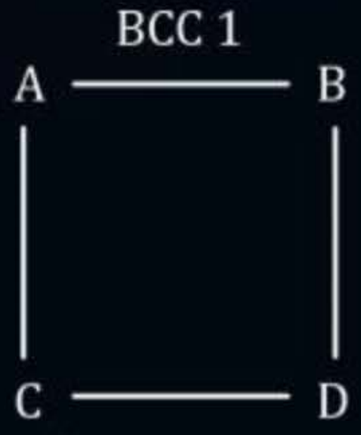


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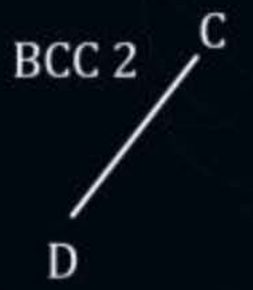
#Q. How many bi-connected components?

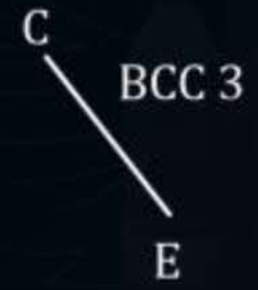
Bi connected graph?→ No (3AP)













5 Bi- CC



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#Q. How many Aps? What?

There are 0 articulation point in G.



- As there are 0 Aps in given graph, if is a Bi- Connected graph.
- Only 1 Bi-cc (graph itself)



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Bi- connected components:

A graph is considered to be bi-connected if it does not have any articulation point in it.

(# AP = 0)

Property:

If a graph is not bi-connected then its maximal sub-graph which is bi-connected is a bi-connected component.



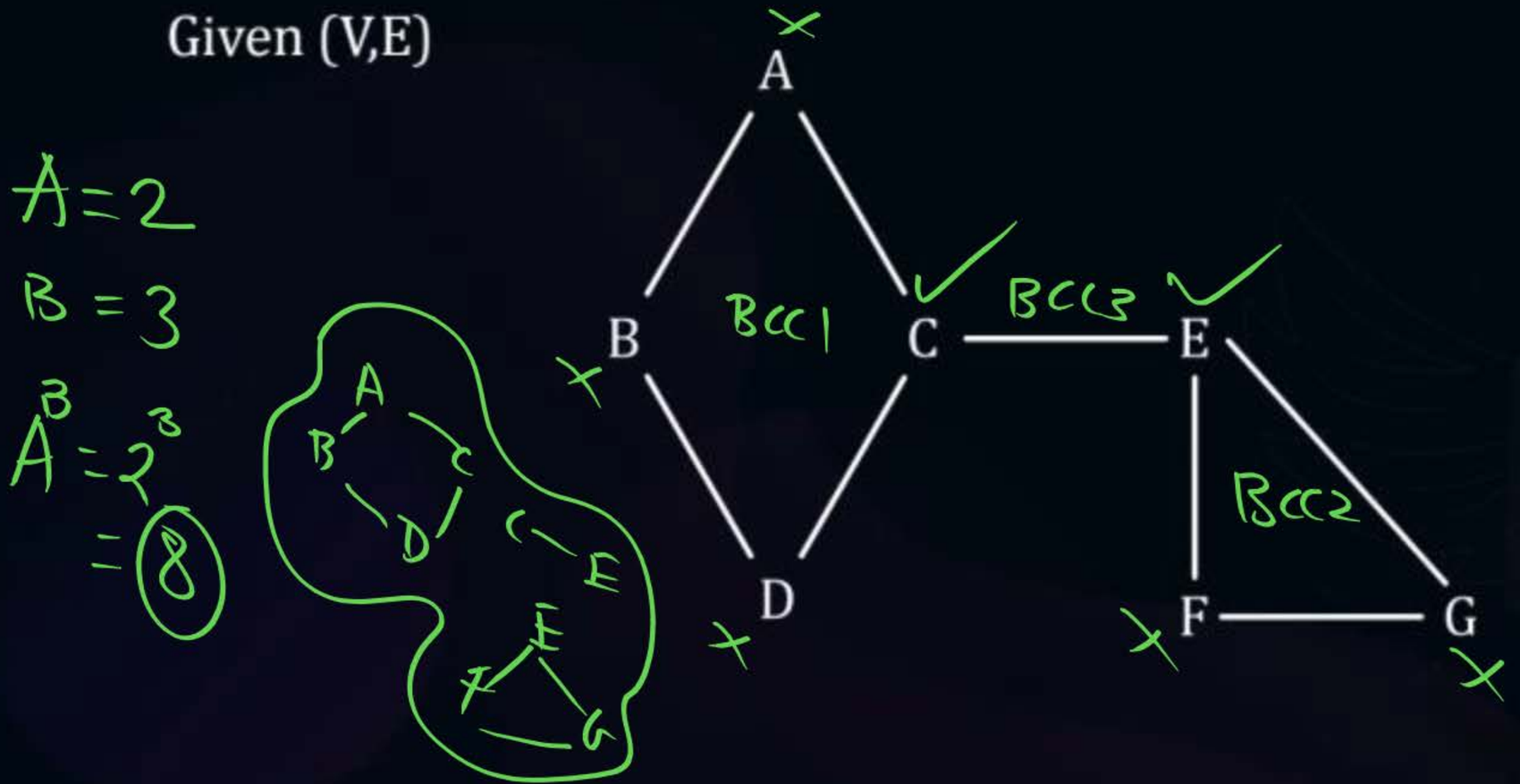
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85%

#Q. Given a graph G, it has A articulation point and B bi-connected components then what is the value of $(A)^B$?

44.6%

Given (V,E)



A = 2

B = 3

$A^B = 2^3 = 8$



THANK - YOU